

Lecture 9: Hypothesis Tests and Confidence Intervals

Last lecture, we provided a few results that will serve as the foundation of inference for OLS. First, we showed that OLS is asymptotically normal, and we provided the large-sample variance of $\hat{\beta}$ for the single regressor case. Next, we showed how to consistently estimate the standard error of $\hat{\beta}$. Today, we will show how to put these two results together in order to test hypotheses and construct confidence intervals.

To prepare us to start doing inference, we will start by doing two reviews. First, we will recap our results from last class. Then, we will review hypothesis testing, borrowing some of the material from our statistics review in Lecture 1. Then, we will show how to combine the two: the fact that OLS is asymptotically normal, combined with our derivation of a robust standard error, allows us to show that the t-statistic is distributed $N(0, 1)$ in large samples, which lets us do hypothesis testing. We will then also show how to construct confidence intervals.

Recap of Asymptotic Normality and Robust Standard Errors

In this lecture, we will continue to use the same three assumptions that we have used throughout, to show that OLS is consistent and asymptotically normal, and later to derive the robust standard error:

Assumption (No Perfect Collinearity). *None of the regressors are perfectly collinear: it is not possible to perfectly predict one of the regressors as a linear combination of the other regressors and a constant.*

Assumption (IID Sampling). *Each observation is drawn independently and from the identical distribution as the other observations.*

Assumption (Orthogonality Condition). *The error term, ε_i , has zero mean $\mathbb{E}[\varepsilon_i] = 0$, and is uncorrelated with the regressors, $\text{Cov}(X_i^k, \varepsilon_i) = 0$ for each k .*

In the last lecture, we showed that, under the above assumptions:

- OLS is asymptotically normal: $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V)$
- For the single regressor case: $V = \frac{\text{Var}((X_i - \mathbb{E}[X_i])\varepsilon_i)}{\text{Var}(X_i)^2}$
- We can construct a robust estimator using the plug-in principle: $\hat{V}_{\text{Robust}} = \frac{\widehat{\text{Var}}((X_i - \bar{X})\hat{\varepsilon}_i)}{\widehat{\text{Var}}(X_i)^2}$
- The robust standard error is consistent: $\hat{V}_{\text{Robust}} \xrightarrow{p} V$

In a moment, we will show how to combine these results and show that the t-test is valid asymptotically. But first, we will do a quick recap of hypothesis testing.

Recap of Hypothesis Testing (Abbreviated from Lecture 1)

We often want to know if it is plausible that a parameter takes a particular value, or lies in some set. To do this, the most common approach is to conduct a **hypothesis test**. A hypothesis test has a few components:

- A **null hypothesis**, H_0 . This is the hypothesis that we are testing about the underlying parameters.
- A **test statistic**, $\tau(X, Y)$, which is a function of the data, which we will denote (X, Y) . Ideally, the test statistic will be low if the data appears consistent with the null, and high if the data does not appear consistent with the null.
- A **critical value**, c , above which the hypothesis test will reject the null.

Hypothesis tests typically can be expressed as a rule:

$$\text{Reject if and only if } \tau(X, Y) > c$$

Note that hypothesis tests can either reject the null hypothesis or “fail to reject” the null hypothesis: we never “accept” the null. Just because the hypothesis test cannot reject the null does not mean that other hypotheses are not also possible.

The **size** of a test is the probability that, conditional on the null being true, the test rejects (a false positive). We will also refer to this as the test’s **level** or **significance level**. A related but distinct concept is the **p-value**, which is the lowest possible size for which the data will still reject the null (we find this by setting the critical value equal to the value of the test statistic in the data, and then figuring out the size associated with that critical value). An equivalent way to think of the p-value is to ask “what is the probability, under the null, that we would observe a test statistic that is at least as large as the one we observed?”

Hypothesis Tests for $\beta = \beta_0$

Suppose that we wish to test the null hypothesis $\beta = \beta_0$. To do this, the most common test statistic is the **t-statistic**. This takes the form

$$\text{t-statistic} = \frac{\hat{\beta} - \beta_0}{\widehat{\text{SE}}(\hat{\beta})}$$

where β_0 is our null hypothesis about β , $\hat{\beta}$ is our estimator, and $\widehat{\text{SE}}(\hat{\beta}) = \sqrt{\hat{V}/N}$ is our standard error (we will use the robust standard error). In practice, we will always take the absolute value of the t-statistic in order to conduct a “two-sided” test, so our actual test statistic will be

$$\tau(X, Y; \beta_0) = \frac{|\hat{\beta} - \beta_0|}{\widehat{\text{SE}}(\hat{\beta})}.$$

If we want to conduct a 5% test, what critical value c should we compare the test-statistic to? To answer this, we can use our results from the last two lectures to show that the t-statistic will converge to a $N(0, 1)$ in large samples.

First, note that from our earlier results, we have:

$$\begin{aligned} \sqrt{N}(\hat{\beta} - \beta) &\xrightarrow{d} N(0, V) \\ \hat{V}_{\text{Robust}} &\xrightarrow{p} V \end{aligned}$$

Then, note that we can write the t-statistic as a function of these two things:

$$\begin{aligned} \text{t-statistic} &= \frac{\hat{\beta} - \beta_0}{\widehat{\text{SE}}(\hat{\beta})} \\ &= \frac{\hat{\beta} - \beta_0}{\sqrt{\hat{V}/N}} \\ &= \frac{\sqrt{N}(\hat{\beta} - \beta_0)}{\sqrt{\hat{V}}} \end{aligned}$$

It then follows (from Slutsky’s Theorem and the Continuous Mapping Theorem) that the t-statistic will converge to a standard normal distribution, if $\beta = \beta_0$:

$$\text{t-statistic} \xrightarrow{d} \frac{N(0, V)}{\sqrt{V}} = N(0, 1)$$

This tells us that, *if the null hypothesis is true*, the t-statistic converges to a standard normal distribution in large samples. This lets us derive the appropriate critical values: we compare the t-statistic to the standard

c	$\Phi(c)$	$\Phi(-c)$	$P(t > c)$
1.645	95%	5%	10%
1.960	97.5%	2.5%	5%
2.576	99.5%	0.5%	1%
3.291	99.95%	0.05%	0.1%

Table 1: Quantiles of the Standard Normal

normal distribution, and reject when the absolute value of the t-statistic is above the appropriately selected critical values. We select these critical values to ensure that, under the null hypothesis $\beta = \beta_0$, the probability of rejecting is equal to our desired significance level. For example, if we want to construct a 5% test, we select a critical value c such that:

$$\begin{aligned}
 P\left(\frac{|\hat{\beta} - \beta_0|}{\widehat{SE}(\hat{\beta})} > c\right) &= 5\% \\
 \implies P(|N(0,1)| > c) &= 5\% \\
 \implies \Phi(c) &= 97.5\% \\
 \implies c &\approx 1.96
 \end{aligned}$$

where $\Phi(c)$ is the cdf of the standard normal distribution, evaluated at c .

Use the Correct Critical Value Why do we select the 97.5th percentile of the standard normal distribution for a 5% test? The important point is that because we are taking the absolute value of the t-statistic (we are performing a “two-sided” or “two-tailed” test), we will reject if the t-statistic is below $-c$ or if it is above c . Since the standard normal is symmetric, each of these events will happen 2.5% of the time under the null. Thus, we select c so that $\Phi(-c) = 2.5\%$ and $\Phi(c) = 97.5\%$. Table 1 provides some useful quantiles of the standard normal, which you can use to conduct tests at various significance levels. **Make sure that when you conduct a 5% test, you use $c = 1.960$ and not $c = 1.645$. This is a very frequent source of error on exams!**

Statistical Power: The Ability to Reject False Hypotheses

Notice that to get the “correct size” of the test (e.g. 5% probability of falsely rejecting a true null), we only care about what the distribution of the test statistic is under the null. However, the hypothesis test has the power to distinguish between true and false null hypotheses because, if the true β is not β_0 , then the test will reject more often than 5%. The probability of rejecting a false null hypothesis is called the **statistical power** of a test. The farther away the true β is from β_0 , and the smaller the standard error, the more powerful our test will be: the more likely we are to be able to reject a false null.

To better understand this, we will work through a simple example. Suppose that we are testing the null hypothesis of $\beta_0 = 3$, and we have a standard error of 2. What we have just shown is that if the true β is in fact equal to three (the null hypothesis), then a 5% test will reject the null 5% of the time:

$$\text{If } \beta = \beta_0 \text{ and } SE=2: P\left(\frac{|\hat{\beta} - \beta_0|}{SE} > 1.96\right) = P\left(\frac{|\hat{\beta} - \beta|}{SE} > 1.96\right) = P(|N(0,1)| > 1.96) = 5\%$$

What if, instead, the null hypothesis were false, and the true β did not equal the null hypothesis? Suppose instead that $\beta = 7 = \beta_0 + 4$. The statistical power is the probability that our test will reject the false null.

We have:

$$\begin{aligned}
 \text{If } \beta = \beta_0 + 4 \text{ and SE}=2: P\left(\frac{|\hat{\beta} - \beta_0|}{\text{SE}} > 1.96\right) &= P\left(\frac{|\hat{\beta} - \beta + 4|}{\text{SE}} > 1.96\right) \\
 &= P\left(\frac{\hat{\beta} - \beta}{\text{SE}} + 2 > 1.96\right) + P\left(\frac{\hat{\beta} - \beta}{\text{SE}} + 2 < -1.96\right) \\
 &= P(N(0,1) > -0.04) + P(N(0,1) < -3.96) \\
 &\approx 51.6\% + 0\% = 51.6\%
 \end{aligned}$$

The probability of rejecting a false null is much higher than the probability of rejecting a true null. The statistical power would increase if the true β were even farther from the null β_0 , and will also increase as the sample size gets larger and the standard error shrinks. As the standard error shrinks, the probability of rejecting a true null remains at the significance level, while the probability of rejecting a false null will approach 100%.

Confidence Intervals for β

We can also use $\hat{\beta}$ and $\widehat{\text{SE}}(\hat{\beta})$ to construct a confidence interval. An $X\%$ confidence interval is an interval with the following property: the true value of β will be in the confidence interval $X\%$ of the time. To be precise, we should think of the confidence interval as a sequence of steps: if we follow those steps to construct our $X\%$ confidence interval, then the true β will be in the interval $X\%$ of the time.

A convenient fact is that if we have hypothesis tests for each null hypothesis β_0 , then we can construct a confidence interval by “inverting” the hypothesis test. For example, if we want a 95% confidence set, then we can test every possible value of β using a 5% test, and include in the confidence set all of the points where the test failed to reject the null. The probability of this procedure excluding the true value of β is 5%, so the probability that it includes the true value is 95%.

For our two-sided test based on the t-statistic, this yields easy to compute confidence intervals. Suppose we wish to construct 95% confidence intervals. The 5% test fails to reject if and only if

$$\begin{aligned}
 \frac{|\hat{\beta} - \beta_0|}{\widehat{\text{SE}}(\hat{\beta})} &\leq 1.96 \\
 \implies |\hat{\beta} - \beta_0| &\leq 1.96 \times \widehat{\text{SE}}(\hat{\beta}) \\
 \implies \beta_0 &\in [\hat{\beta} - 1.96 \times \widehat{\text{SE}}(\hat{\beta}), \hat{\beta} + 1.96 \times \widehat{\text{SE}}(\hat{\beta})]
 \end{aligned}$$

So, in general we construct our confidence intervals by taking the appropriate c and constructing the confidence interval from $\hat{\beta} \pm c \cdot \widehat{\text{SE}}(\hat{\beta})$.

Exercise 1. Suppose we estimate a regression and find $\hat{\beta} = 0.4$ and $\widehat{\text{SE}}(\hat{\beta}) = 0.15$. Test the null hypothesis $\beta_0 = 0$ using a two-sided test, at the 5% significance level. State the test statistic, critical value, and the conclusion of the test.

Exercise 2. Continuing with the setting of the previous exercise, construct a 95% confidence interval for β . Now, construct a 90% confidence interval and a 99% confidence interval.

Exercise 3. Suppose that we estimate a 95% confidence interval for β of $[0.3, 1.3]$. Are we able to reject the null $\beta_0 = 0$ at the 5% level? What about the null hypothesis $\beta_0 = 0.5$?

Exercise 4. In the setting of the previous exercise, assuming that the confidence interval was constructed in the standard manner, what is $\hat{\beta}$ and what is $\widehat{\text{SE}}(\hat{\beta})$?